TASK TELANGANA · CLOUD SOFTWARE DEVELOPER TRAINING

Mini Project 1: Deploying a Cloud Portfolio on AWS EC2

Step-by-Step Hands-on Guide: Ubuntu 24.04 LTS + Nginx + Git + GitHub

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PROJECT OBJECTIVE: Every student will launch an enterprise-grade Ubuntu 24.04 LTS instance on AWS Free Tier, configure the high-performance Nginx web server, clone their developer portfolio from GitHub, fix Linux permissions (www-data), and test the live public URL.

1. System Architecture Flow

Internet (Recruiters) ➔ Security Group (Port 80) ➔ Ubuntu EC2 ➔ Nginx ➔ /var/www/html (GitHub Portfolio)

2. Step-by-Step Execution Playbook

Step #	Action Item	Terminal Commands / Console Actions
1	**Launch EC2**	• Name: gauri-portfolio • AMI: **Ubuntu Server 24.04 LTS** • Type: t2.micro or t3.micro (Free Tier) • Key Pair: Select your key pair • Auto-assign Public IP: **Enable** • Security Group: **SSH Port 22 (My IP)** + **HTTP Port 80 (0.0.0.0/0)**
2	**Connect**	AWS Console ➔ Select Instance ➔ Click **Connect ➔ EC2 Instance Connect ➔** Connect as ubuntu.
3	**Update OS**	sudo apt update sudo apt upgrade -y
4	**Install Tools**	sudo apt install git nginx -y Check status: sudo systemctl status nginx (Active: running)
5	**Test Nginx**	Open in browser: http://<YOUR_EC2_PUBLIC_IP> Verify the default 'Welcome to nginx!' page loads.
6	**Deploy GitHub Portfolio**	cd /var/www/html sudo rm -f index.nginx-debian.html sudo git clone https://github.com/gauri-abc/gauri-yadav-cloud-portfolio.git . *(Or clone your own GitHub repository URL)*
7	**Fix Permissions**	sudo chown -R www-data:www-data /var/www/html sudo chmod -R 755 /var/www/html
8	**Restart Nginx**	sudo systemctl restart nginx sudo systemctl enable nginx
9	**Final Verification**	Hard-refresh browser: http://<YOUR_EC2_PUBLIC_IP> Your personal portfolio website is now live on AWS!

3. Common Pitfalls & Troubleshooting

- **Browser Timeout:** Verify Security Group has Inbound HTTP Port 80 open to 0.0.0.0/0.
- **403 Forbidden:** Re-run permissions command: sudo chown -R www-data:www-data /var/www/html && sudo chmod -R 755 /var/www/html.
- **Still seeing Nginx default page:** Ensure index.nginx-debian.html was removed before cloning.
- **HTTPS redirection error:** Type http:// explicitly in your browser, as free EC2 instances do not have an SSL certificate installed by default.